Supporting Information for Diffusion of Colloidal Rods in Corrugated Channels

Xiang Yang,[1] Qian Zhu,[1] Chang Liu,[1] Wei Wang,[2] Yunyun Li,[3] Peter Hänggi,[4,5] Fabio Marchesoni,[3,6] and H. P. Zhang[1,7]

[1]*School of Physics and Astronomy and Institute of Natural Sciences, Shanghai Jiao Tong University, Shanghai, China*

[2]*School of Material Science and Engineering, Harbin Institute of Technology, Shenzhen Graduate School, Shenzhen,China*

[3]*Center for Phononics and Thermal Energy Science, School of Physics Science and Engineering, Tongji University, Shanghai, China*

[4]*Institut für Physik, Universität Augsburg, D-86135 Augsburg, Germany*

[5]*Nanosystems Initiative Munich, Schellingstrasse 4, D-80799 München, Germany*

[6]*Dipartimento di Fisica, Università di Camerino, I-62032 Camerino, Italy*

[7]*Collaborative Innovation Center of Advanced Microstructures, Nanjing, China*

(Dated: November 24, 2018)

Abstract

This document contains supporting information for the article entitled "Diffusion of Colloidal Rods in Corrugated Channels"

I. MATERIAL AND METHOD

A. Colloidal rod fabrication

Gold microrods were synthesized by electrodeposition in alumina templates following a procedure adapted from an earlier study [1]. Porous AAO membranes (Whatman) with a nominal pore diameter of 200 nm were used. Before electrodeposition, a 200 - 300 nm layer of silver was thermally evaporated on one side of the membrane as the working electrode. The membrane was then assembled into an electrochemical cell with the pore openings immersed in the metal plating solution. Silver plating solution (Alfa Aesar) and homemade gold solution (gold content 28.7 g/L) were used. In a typical experiment, a 5 - 10 μm layer of silver was first electro-deposited into the pores at -5 mA/cm^2, followed by gold at -0.2 mA/cm^2, the amount of which was controlled by monitoring the charge flow. The pores' silver filling and the membrane were then dissolved, respectively in HNO_3 and NaOH solutions, and the released gold rods were cleaned in distilled water. In the final step of the fabrication process, a thin layer (50 nm) of iron was then thermally evaporated by electron beam in a vacuum onto the sides of the gold rods (e-beam evaporator HHV TF500). Rod width is about 300 nm and we selected straight rods with length from 1.6 μm to 3.2 μm for our experiment.

B. Local diffusivity measurements

Confining boundaries cause the normal diffusivity of the rods to spatially vary inside the channel. We measured the local diffusivity in the rod body frame $\mathbb{D}_{IJ}(x, y, \theta)$ $(I,\ J = X, Y, \theta)$ via the displacement covariance matrix. As illustrated in Fig. S1(a) the elements of the covariance matrix at a given sample point [marked by a cross in Fig. S1(c)] can be fitted by linear functions of the elapsed time, δt, with slope proportional to the relevant diffusivity element, $\langle \delta I \delta J \rangle = 2\mathbb{D}_{IJ}(x, y, \theta)\delta t$ (normal diffusion). All non-diagonal cross-covariances, such as $\langle \delta X\,(\delta t)\,\delta Y\,(\delta t) \rangle$ shown in Fig. S1(a), are much smaller than the variances, which means that the diffusivity matrix $\mathbb{D}_{IJ}(x, y, \theta)$ is dominated by its diagonal elements. In Fig. S1(b) all displacements after a time interval δt =0.2 s exhibit a normal Gaussian distribution, which suggests that our choice of δt was appropriate. We carried out local diffusivity measurements at 200 points uniformly distributed in the configuration space

x, y, θ; then used such measurements to interpolate the diffusivity matrix $\mathbb{D}_{IJ}(x, y, \theta)$ over the entire parameter space. Measured $\mathbb{D}_{XX}$, $\mathbb{D}_{YY}$, $\mathbb{D}_{\theta\theta}$ for three tilting angles are depicted in Figs. S1(c)-(e) – see Fig. S3(a)-(b) for the corresponding numerical results.

C. Finite-element simulations

We complemented our experimental data for the local diffusivity with finite-element calculations. COMSOL Multiphysics v5.2 was used to compute the drag force on a moving rod in a channel. The Stokes equations were solved. In doing so, no-slip boundary conditions were imposed on the side walls, floor and ceiling, while open boundary conditions were adopted for the channel openings. The geometry of the channel was set to reproduce the inner channel boundary in the experiments. We used about half a million elements in the simulation to ensure convergence. The same numerical method was followed to solve the problem of a sphere moving in a long cylinder; the computed drag force on the sphere deviates less than 3% from the analytical predictions [2], which can be regarded as an estimate of the achieved numerical accuracy.

The rod dwells at different locations and orientations, (x, y, θ), in a horizontal plane at about H_{rod} =0.4 μm above the floor. We estimated H_{rod} by balancing the rod gravity (about 0.025 pN), buoyancy and the electrostatic repulsion between the rod and the channel walls, for which we assume a Debye length of 60 nm and surface potentials of 40 mV [3, 4]. As shown in Supporting Movie S1.mp4, the rods exhibit little out-of-plane motion; therefore, in our simulations and experimental data analysis we assumed that they stay in the same horizontal plane at all times.

For each (x, y, θ), we dragged the rod with constant velocity $v = 1\mu$m/s in the x or y direction, and measured the corresponding drag force f_x and f_y in each case. We calculated the hydrodynamic friction coefficient matrix $\gamma = \{\gamma_{ij}\}$, $(i, j = x, y)$ through the relation $\vec{f} = \gamma\vec{v}$. Then, the diffusion matrix was derived from the γ matrix via the fluctuation-dissipation theorem[5].

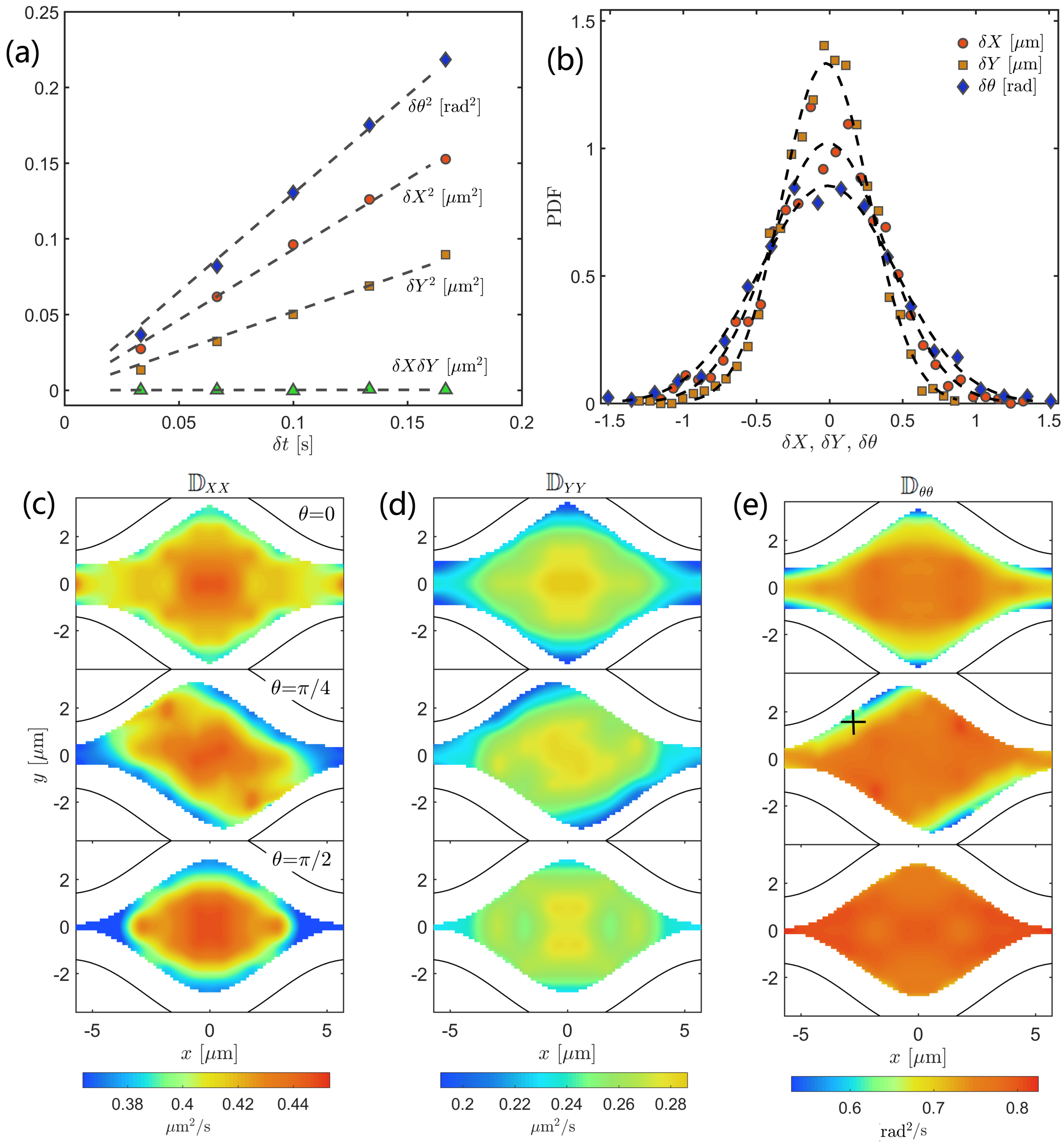

Figure S1. (a) Covariances of linear and angular displacements, $\langle \delta X^2 \rangle$, $\langle \delta Y^2 \rangle$, $\langle \delta X \delta Y \rangle$ and $\langle \delta\theta^2 \rangle$, versus elapsed time δt at coordinates: (x=-2.5 μm, y=1.6 μm, θ=$\pi/4$), marked by a cross in the middle panel of (e). Linear fits were used to extract the corresponding diffusivity matrix element. (b) Probability distributions of δX, δY, and $\delta\theta$ at $\delta t = 0.2$ s fitted by Gaussian functions (dashed curves) with corresponding variance from panel (a). (c)-(e) Diagonal terms of the local diffusivity in the body frame (c) $\mathbb{D}_{XX}(x, y, \theta)$, (d) $\mathbb{D}_{YY}(x, y, \theta)$ and (e) $\mathbb{D}_{\theta\theta}(x, y, \theta)$ are measured at $\theta = 0,\ \pi/4$ and $\pi/2$. Data were taken in a tall channel with $h_n = 1.4\mu$m and $l_X = 1\mu$m; the channel's inner boundaries $\pm h(x)$ are marked by solid lines.

D. Brownian Dynamics (BD) simulations

Because the inertial forces are negligible with respect to the viscous forces, we used overdamped Langevin equations to describe the evolution of rod coordinates and orientation, denoted by the vector $\vec{s} = (x, y, \theta)$:

$$\frac{d\vec{s}}{dt} = -\mathbb{R}\frac{\mathbb{D}(\vec{s})}{k_B T}\vec{F}(\vec{s}) + \mathbb{R}\vec{\xi}(t), \tag{S1}$$

where $\mathbb{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is the transformation matrix from the body to the laboratory frame. The diffusion matrix $\mathbb{D}(\vec{s})$ in the body frame contains diagonal elements only, $\mathbb{D}_{XX}(\vec{s})$, $\mathbb{D}_{YY}(\vec{s})$, $\mathbb{D}_{\theta\theta}(\vec{s})$, which are provided either by experimental measurements or finite-element simulations. Interaction force and torque between rod and boundary, $\vec{F}(\vec{s})$, are computed in the body frame (see below). Fluctuations in the body frame are independent Gaussian white noises $\vec{\xi}(t)$, with $\langle \xi_I(t) \rangle = 0$ and $\langle \xi_I(t+\tau)\xi_J(t) \rangle = 2\mathbb{D}_{IJ}(\vec{s})\delta(\tau)\delta_{IJ}$, where $I(J)$ represents X, Y or θ, as appropriate.

We computed the rod-wall interaction in the following way. As illustrated in Fig. S2, the channel boundary and the rod are represented by a string of particles: the l-th boundary particle is denoted by $\vec{r_l} = (x_l, y_l)$ and the k-th rod particle by $\vec{r_k}$. These particles interact with each other via the truncated Lennard-Jones(LJ) potential $U(|\vec{r_l} - \vec{r_k}|)$,

$$U(r) = \begin{cases} U_{LJ}(r) - U_{LJ}(1.12\sigma), \ r \leq 1.12\sigma \\ 0, \ r > 1.12\sigma \end{cases} \tag{S2}$$

$$U_{LJ}(r) = 4\varepsilon[(\frac{\sigma}{r})^{12} - (\frac{\sigma}{r})^{6}], \tag{S3}$$

where ε and σ are the strength and characteristic length of the potential respectively. Total force and torque acting on the rod are combined in the vector

$$\vec{F}(\vec{s}) = \begin{pmatrix} -\sum\limits_{k,l} \nabla_X U(|\vec{r_l} - \vec{r_k}|) \\ -\sum\limits_{k,l} \nabla_Y U(|\vec{r_l} - \vec{r_k}|) \\ -\sum\limits_{k,l} |\vec{r_l} - \vec{r}| \nabla_Y U(|\vec{r_l} - \vec{r_k}|) \end{pmatrix},$$

where $\vec{r} = (x, y)$ denotes the center of the rod.

We had recourse to Euler's method to discretize the Langevin equations with time step dt. For thermodynamic consistency, we used the transport (also known as kinetic or isothermal) convention [6–8], and a predictor-corrector scheme [9] to compute the post-point value, $\vec{s}(t + dt)$. We first predict the post-point value from the information at $\vec{s}(t)$

$$\vec{s}^*(t + dt) = \vec{s}(t) + \mathbb{R}\frac{\mathbb{D}(\vec{s}(t))}{k_B T}\vec{F}(\vec{s}(t))dt + \mathbb{R}\sqrt{2\mathbb{D}(\vec{s}(t))dt}\vec{\eta}(t), \tag{S4}$$

and then improve it in a corrector step

$$\vec{s}(t + dt) = \vec{s}(t) + \mathbb{R}\frac{\mathbb{D}(\vec{s}^*(t + dt))}{k_B T}\vec{F}(\vec{s}^*(t + dt))dt + \mathbb{R}\sqrt{2\mathbb{D}(\vec{s}^*(t + dt))dt}\vec{\eta}(t) \tag{S5}$$

where the vector $\vec{\eta}(t)$ consists of three independent Gaussian random numbers with zero mean and unit variance. Our numerical scheme produces uniformly distributed rods in the accessible space as in the experiments, which validates the scheme itself.

In our simulations we used the time step $dt = 0.2$ ms, potential parameters $\sigma = 0.1\mu$m and $\varepsilon = 2k_BT$, and located the wall particles $\vec{r}_k$, so that the space accessible to the diffusing rod was the same as in the experiments. Upon rescaling length, time, and energy respectively by L, $L^2/D_{\text{ave}}(x = 0)$ [see Eq. (3) in the main text], and k_BT, the dimensionless simulation parameters became $\sigma = 8 \times 10^{-3}$, $dt = 4 \times 10^{-7}$, and $\varepsilon = 2$. Particle trajectories from simulation can be analyzed by the same technique (see main text) adopted for their experimental counterparts to determine the relevant MFPTs.

II. DISCUSSION

A. Local diffusivity measurements for finite-element simulations

Typical finite-element calculation results for $\mathbb{D}_{XX}(x, y, \theta)$ and $\mathbb{D}_{YY}(x, y, \theta)$ are plotted in Fig. S3(a)-(b). Upon integrating $\mathbb{D}_{XX}$ and $\mathbb{D}_{YY}$, as prescribed in Eq. (3), we obtained the local diffusivity along the channel direction, $D_{\text{ave}}(x)$, plotted in Fig. S3(c). These numerical results agree with their experimental counterpart of Fig. S1. For example, both experiments and finite-element calculations suggest that for long rods $D_{\text{ave}}(x)$ is largest at the center of the neck. We systematically investigated this phenomenon in Fig. S3(d) over a wide range

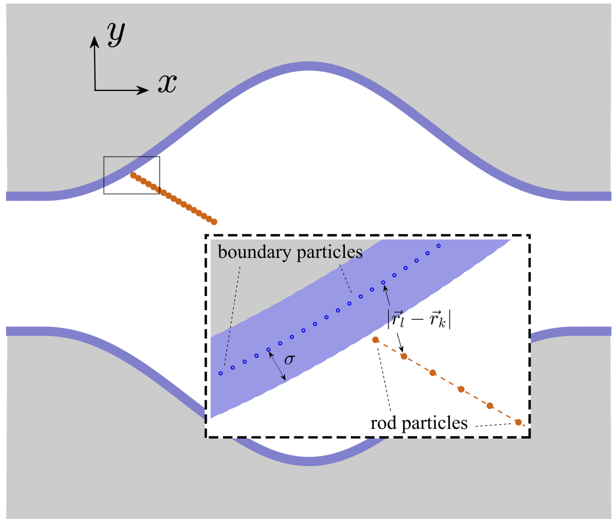

Figure S2. Schematic diagram of the channel and rod model used to compute the rod-wall interactions in BD simulations. Fixed particles (located at $\vec{r}_k$, separated by 0.25σ) mimicking the walls are shown in blue and those mimicking the diffusing rod (located at $\vec{r}_l$, separated by 0.8σ) in orange. The quantity σ is the characteristic length of the LJ potential (see text). Periodic boundary conditions are imposed at the left and right openings.

of the channel width, h_n, and rod length, l_X. For rods with $l_X > 1.2\,\mu$m, $D_{\mathrm{ave}}\,(x = L/2)$ is greater than $D_{\mathrm{ave}}\,(x = 0)$. As a comparison, we calculate the rod translational diffusivity at the center of the channel cells with $\theta = 0$; namely, $D_0 = [\mathbb{D}_{XX}(0,0,0) + \mathbb{D}_{YY}(0,0,0)]/2$ and in the unbounded space, $D_{\mathrm{bulk}} = (\mathbb{D}_{XX} + \mathbb{D}_{YY})/2$. Numerical values for these quantities are shown in the caption of Fig. S3.

B. Diffusion time scales and the generalized FJ approach

With the typical geometric parameters (l_X =1 μm, h_n =1.4 μm, h_w =4.2 μm and L =12 μm) adopted in our experiments, we estimated the characteristic diffusion times in the x (channel) direction, $\tau_x = L^2/2D_0 \simeq 200$ s; in the y direction, $\tau_y = h^2/2D_0 \simeq 23$ s (cell center) and 3 s(neck); and for the rod's rotation $\tau_\theta = (\pi/2)^2/2\mathbb{D}_{\theta\theta} \simeq 1.5$ s. Therefore, relaxation in the y and θ directions are much faster than in the channel direction, which justifies implementing the FJ approach. We also note that $\tau_\theta \ll \tau_y$ in most space except in the neck region.

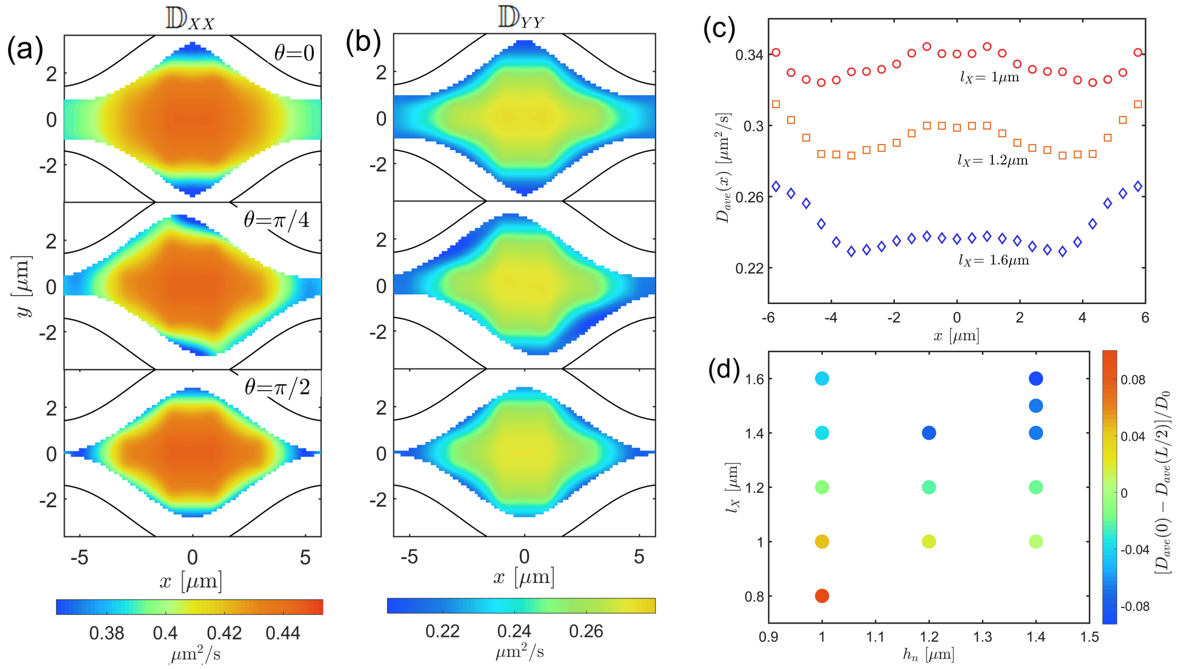

Figure S3. Numerical diffusivity results from finite-element calculation. Local diffusivity in the body frame (a) $\mathbb{D}_{XX}(x, y, \theta)$ and (b) $\mathbb{D}_{YY}(x, y, \theta)$ computed for $\theta = 0$, $\pi/4$, $\pi/2$. (c) Average local diffusivity $D_{\text{ave}}(x)$ vs. x for three rods of different length, for which the diffusivity at the center of the channel was D_0 = 0.36,0.34 and 0.24 μm^2/s, to be compared with the diffusivity in the unbounded space, D_{bulk} = 0.47, 0.42 and 0.34 μm^2/s (see supporting text for definitions). (d) Values of $[D_{\text{ave}}(0) - D_{\text{ave}}(L/2)]/D_0$ in the h_n-l_X plane. Data were taken in a tall channel (H = 2.0 μm, $\alpha = 1$) with h_n =1.4 μm, see Figs. S1(c)-(d) and 3(a) for the corresponding experimental results.

C. Reguera-Rubì approximation for effective diffusivity

According to the FJ approach, the adiabatic elimination of the transverse coordinates leads to entropic corrections to the effective local diffusivity. For a 2D channel with confining boundaries at $\pm g(x)$, Reguera and Rubí (RR) proposed the following expression for the effective diffusivity: $D(x) = D_0[1 + g'(x)^2]^{-\frac{1}{3}}$, if $g'(x) < 1$ [10]. RR also provided a formula for 3D axisymmetric channels with spatial d.o.f.'s only. However, their formula does not apply to a generic 3D channel. We approximated the reconstructed 3D channel [Fig. 2(b)] to a quasi-2D one with half-width $G(x)$ and applied the improved RR approximation, $D(x) = D_{\text{ave}}(x)\,[1 + G'(x)^2]^{-\frac{1}{3}}$. As shown in Figs. S4(a)-(b), such an approximation raises the

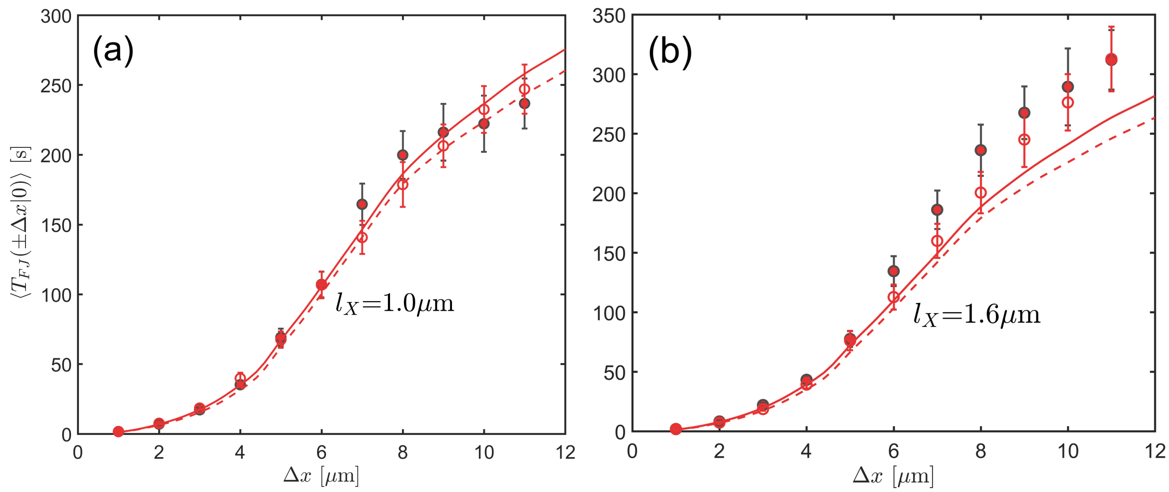

Figure S4. MFPT from experiments (filled symbols), BD simulations (open symbols) and FJ predictions with (solid curves) or without the improved RR correction (dashed curves). Data were taken in tall channels (H =2 μm, α = 1) with h_n =1.4 μm for two different rod half-lengths, (a) l_X = 1.0 and (b) 1.6μm. In the BD simulations we made use of the experimentally measured diffusivity matrix.

theoretical estimates by some 5% (solid versus dashed curves). For a short rod, l_X =1 μm, theoretical predictions are consistent with both experimental and simulation results. For a long rod, l_X =1.6 μm, however, the RR formula underestimates both the experimental and simulation results by ∼15% – note that the experimental and numerical data sets agree with each other for all values of l_X. This may be understood by noticing that the reconstructed 3D channel for a long rod, see Figs. 2 (b)-(c), exhibits large variations along the θ axis and, therefore, cannot be accurately represented by an averaged quasi-2D channel.

D. Dependence of the MFPT on channel neck width and rod length

The width of the channel's neck, h_n, and the length of the rod, l_X, affect the MFPT by changing both the entropic barrier and the effective diffusivity in Eq. (5). To clarify their combined effect, we denote $\langle T_{FJ}(\pm\Delta x|0)\rangle$ by $T(\omega;\Delta x)$, where ω represents either h_n or l_X, as appropriate, and rewrite Eq. (5) as

$$T(\omega;\Delta x) = \int_0^{\Delta x} \lambda(\omega;\eta)\frac{1}{D_{\rm ave}(\omega;\eta)}d\eta, \tag{S6}$$

where

$$\lambda(\omega;x) = \frac{[1+G'(\omega;x)^2]^{\frac{1}{3}}}{G(\omega;x)} \int_0^x G(\omega;\xi)d\xi$$

depends only on the geometric properties of the system. To further advance our analysis, we assume a weak x dependence of the local diffusivity $D_{\rm ave}(\omega;x)$. This assumption allows us to replace $D_{\rm ave}(\omega;\eta)$ in Eq. (S6) with its average, $\bar{D}(\omega) \equiv \frac{1}{L}\int_0^L D_{\rm ave}(\omega;\eta)d\eta$. Accordingly, $T(\omega;\Delta x)$ is approximated by

$$T_0(\omega;\Delta x) = \frac{1}{\bar{D}(\omega)} \int_0^{\Delta x} \lambda(\omega;\eta)d\eta.$$

We now take the partial derivative of the MFPT at $\Delta x = L/2$ [see Fig. 4(b)], that is

$$\begin{aligned} \frac{\partial T_0(\omega;L/2)}{\partial \omega} &= \frac{1}{\bar{D}(\omega)} \int_0^{L/2} \frac{\partial \lambda(\omega;\eta)}{\partial \omega} d\eta + \frac{\partial}{\partial \omega}\bar{D}(\omega)^{-1} \int_0^{L/2} \lambda(\omega;\eta)d\eta \\ &= T_0(\omega;\ L/2)\left(\frac{\int_0^{L/2}\frac{\partial\lambda(\omega;\eta)}{\partial\omega}d\eta}{\int_0^{L/2}\lambda(\omega;\eta)d\eta} + \bar{D}(\omega)\frac{\partial}{\partial\omega}\bar{D}(\omega)^{-1}\right). \end{aligned} \tag{S7}$$

From this equation we immediately realize that a change in the system parameter ω generates two contributions, an entropic term, $e_\omega(\omega) = \int_0^{L/2} \frac{\partial\lambda(\omega;\eta)}{\partial\omega}d\eta / \int_0^{L/2}\lambda(\omega;\eta)d\eta$, and a diffusion term, $u_\omega(\omega) = \bar{D}(\omega)\frac{\partial}{\partial\omega}\bar{D}(\omega)^{-1}$.

As h_n and l_X change proportionally by the amounts dl_X and dh_n, i.e. $dl_X/dh_n = l_X/h_n$, the total change of T_0 reads

$$dT_0 = T_0[u_{l_X} + e_{l_X} + \frac{h_n}{l_X}(u_{h_n} + e_{h_n})]dl_X. \tag{S8}$$

We computed each of these terms numerically in a tall channel with different l_X and h_n. For the diffusive terms we used diffusivity matrices from finite-element calculations. All four terms of Eq.(S8) and their sum are plotted in Fig. S5. An increase of the rod length l_X leads to less available configuration space and higher entropic barriers [see Fig. 2 (c)], that causes the MFPT to increase. Vice versa, increasing h_n reduces the height of the entropic barrier and thus decreases the MFPT. This corresponds to positive e_{l_X} and negative e_{h_n} values in Fig. S5(a). A longer rod is naturally characterized by smaller diffusivity [see Fig. 3(a)] and longer MFPT, which is consistent with a positive value of u_{l_X} in Fig. S5(b). Changes in the neck width, h_n, has a weak impact on the effective diffusivity, as proven by the small

u_{h_n} values in Fig. S5(b). Adding up all four terms yields a vanishing MFPT increment, $dT_0 \sim 0$, for all data points in Fig. S5(c), except the two points encircled by a dashed ellipse. These two outliers represent the case of two short rods, $l_X = 0.6$, and 0.5μm in an extremely narrow channel with h_n =0.6 μm (i.e., comparable with the rod width $2l_Y$ =0.3 μm). Under these conditions, a change in h_n can lead to large values of e_{h_n} and u_{h_n}, also marked by dashed ellipses in Figs. S5(a)-(b).

III. DESCRIPTION OF SUPPORTING VIDEO

Supporting video (S1.mp4) shows a typical rod trajectory plotted on an optical image of the channel. Rod orientation is reported according to a color-code. Some of these data have been plotted in Fig. 1(b).

---

[1] W. Wang, T.-Y. Chiang, D. Velegol, and T. E. Mallouk, J. Am. Chem. Soc. **135**, 10557 (2013).

[2] K. Misiunas, S. Pagliara, E. Lauga, J. R. Lister, and U. F. Keyser, Phys. Rev. Lett. **115**, 038301 (2015).

[3] T. Y. Chiang and D. Velegol, Langmuir **30**, 2600 (2014).

[4] J. Israelachvili, *Intermolecular and Surface Forces* (Academic Press, 2011).

[5] R. Kubo, Rep. Prog. Phys. **29**, 255 (1966).

[6] P. Hänggi, Helv. Phys. Acta **51**, 183 (1978).

[7] I. M. Sokolov, Chem. Phys. **375**, 359 (2010).

[8] O. Farago and N. Gronbech-Jensen, Phys. Rev. E **89**, 013301 (2014).

[9] N. Bruti-Liberati and E. Platen, Stochastics and Dynamics **08**, 561 (2008).

[10] D. Reguera, G. Schmid, P. S. Burada, J. M. Rubí, P. Reimann, and P. Hänggi, Phys. Rev. Lett. **96**, 130603 (2006).

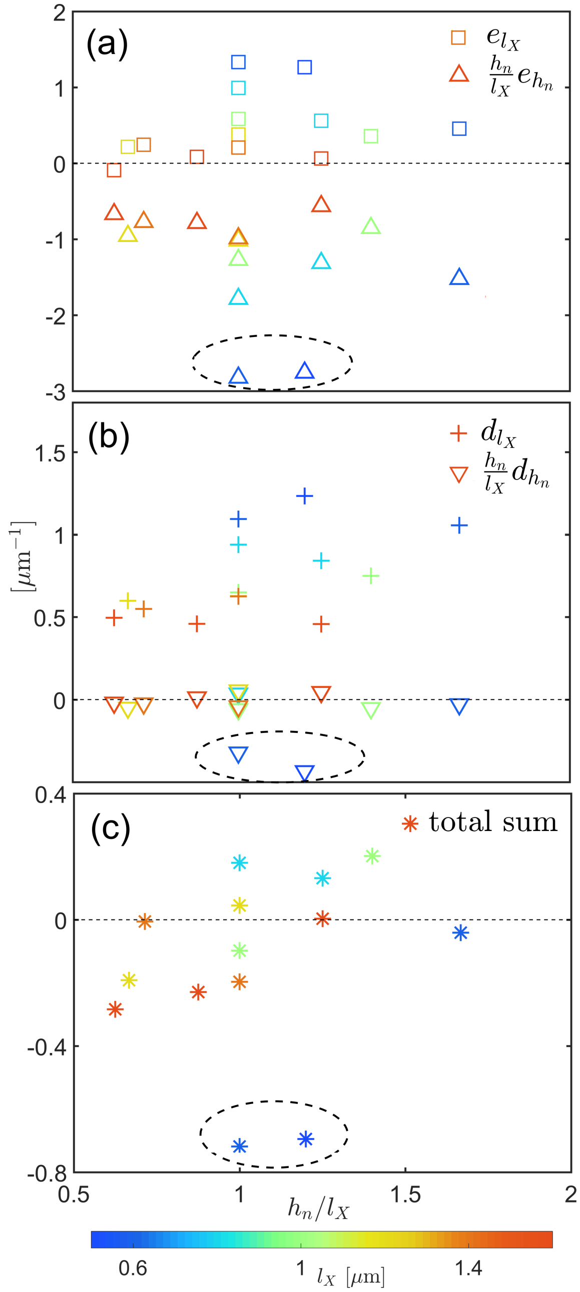

Figure S5. Illustration of Eq. (S8): (a) entropic terms e_{l_X} and $(h_n/l_X)e_{h_n}$, (b) diffusive terms u_{l_X} and $(h_n/l_X)u_{h_n}$, and (c) sum of all four terms plotted vs. h_n/l_X. Data points from the two shortest rods are encircled by dashed ellipses (see text). All data points are color coded according to the rod half-length, l_X.